Tutorial 5: FINA2205 Portfolio Theory and Asset Pricing 1

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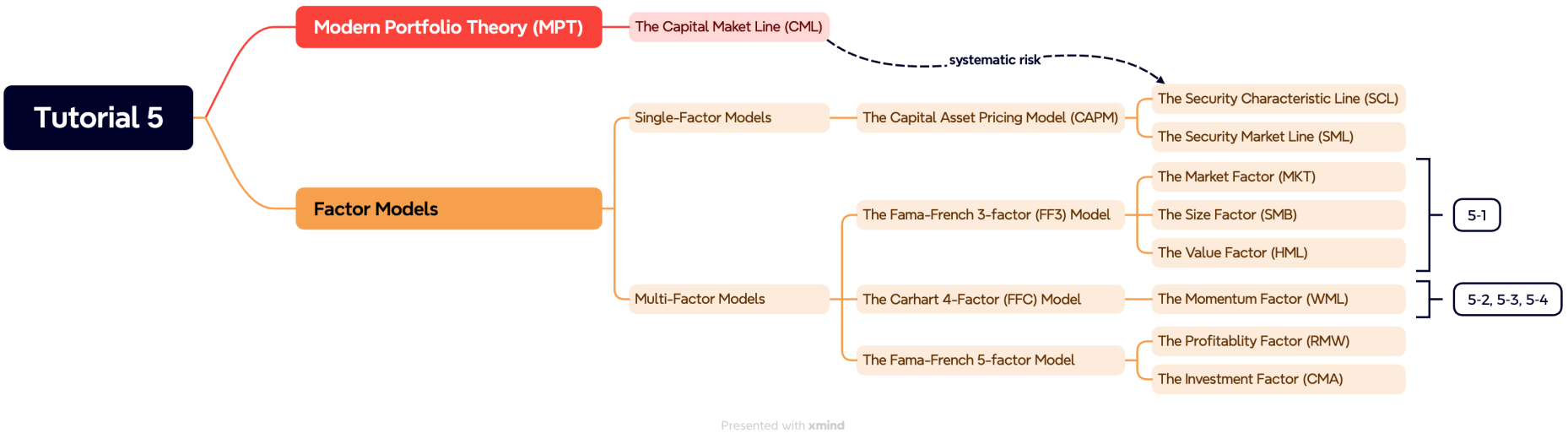
Agenda

- 4 problems

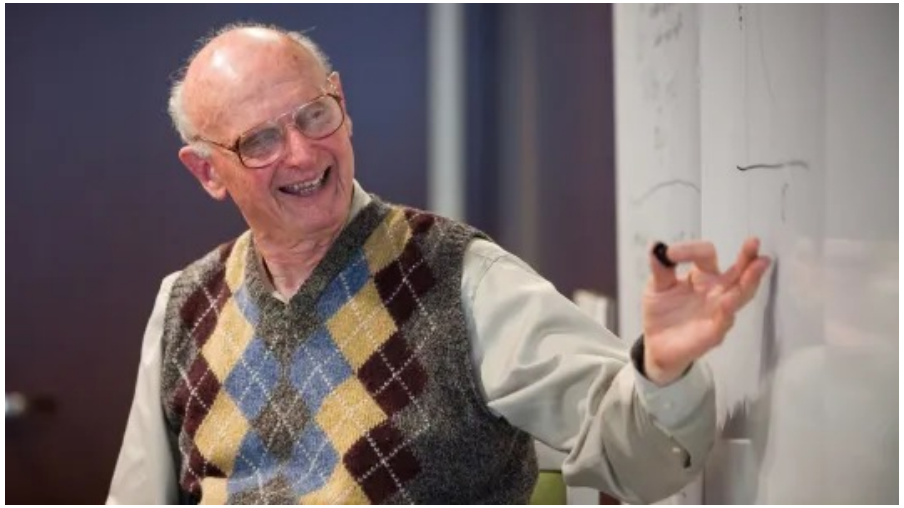


Recap: Modern Portfolio Theory (MPT)

- Example: street vendor: umbrellas vs. sunglasses.
- Diversification: "*Don't put all your financial eggs into one basket*".
- By combining stocks that face different risks and do not move in perfect sync, a portion of the tota risk is naturally cancelled out.
- **The problem**: How do we mathematically prove which combination of baskets is the absolute best?

Recap: Modern Portfolio Theory (MPT)

- **Harry Markowitz** (1952) solved this asset allocation puzzle, winning the 1990 Nobel Prize in Economics for his foundational work on MPT.
- Core idea: Don't judge a stock just by its own ups and downs. Judge it by how movements interact with the rest of your investments.



Recap: Modern Portfolio Theory (MPT)

A portfolio of two securities

- Let x_1 be the weight invested in asset A, and x_2 the weight invested in asset B.
- The portfolio expected return is:

$$E(P) = x_1 E(A) + x_2 E(B)$$

- The portfolio risk (σ_P) is:

$$\sigma_P = \sqrt{x_1^2\sigma_A^2 + 2x_1x_2\sigma_{AB} + x_2^2\sigma_B^2} = \sqrt{x_1^2\sigma_A^2 + 2x_1x_2\sigma_A\sigma_B\rho_{AB} + x_2^2\sigma_B^2}$$

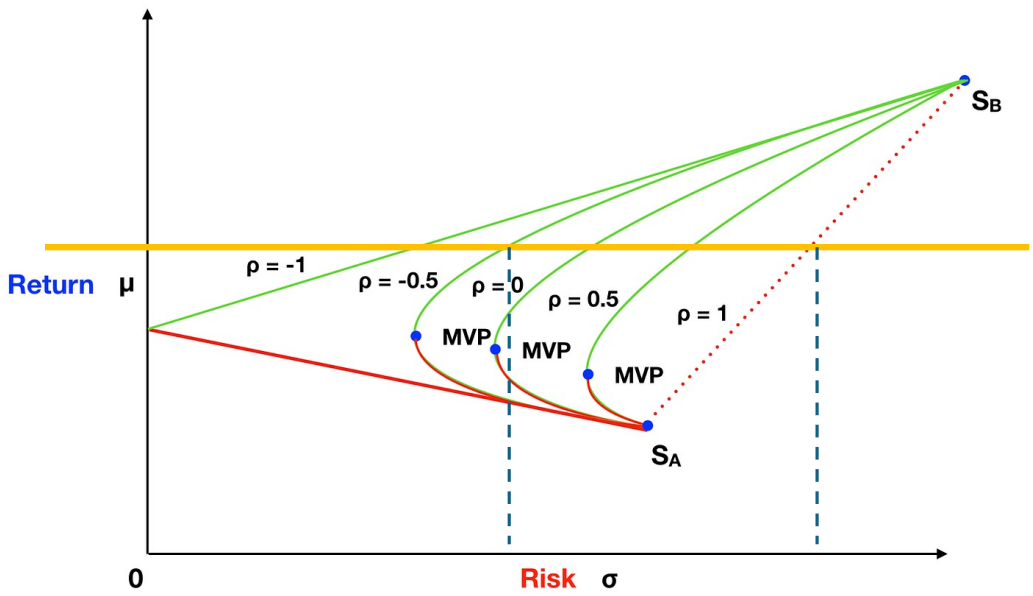
covariance

correlation $-1 \leq \rho \leq 1$

Recap: Modern Portfolio Theory (MPT)

2-asset math

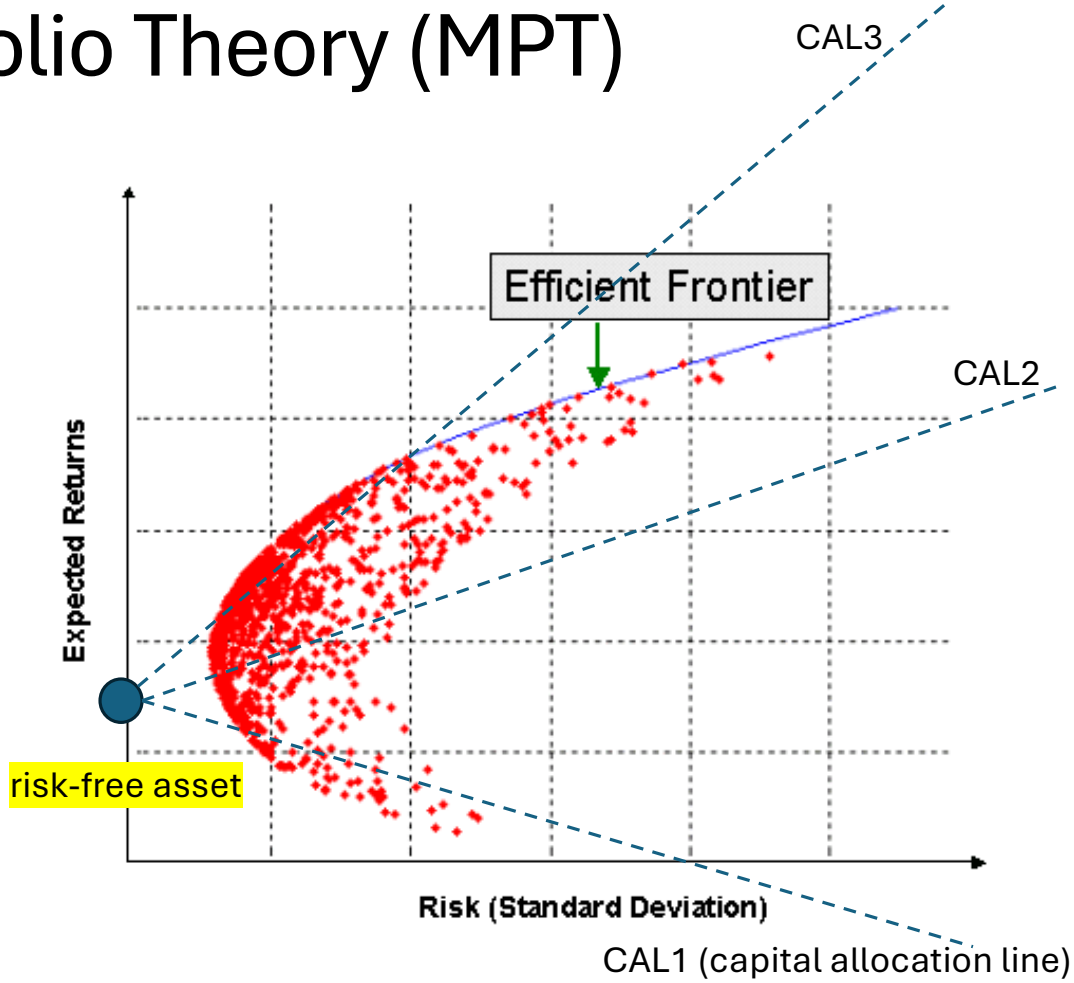
- S_A (lower risk, lower return)
- S_B (higher risk, higher return)
- If $\rho = 1$, two stocks move in perfect sync. There is zero diversification benefit.
- If $\rho < 1$, visually, the line starts "bowing" to the left.



$\sigma_2 < \sigma_1$

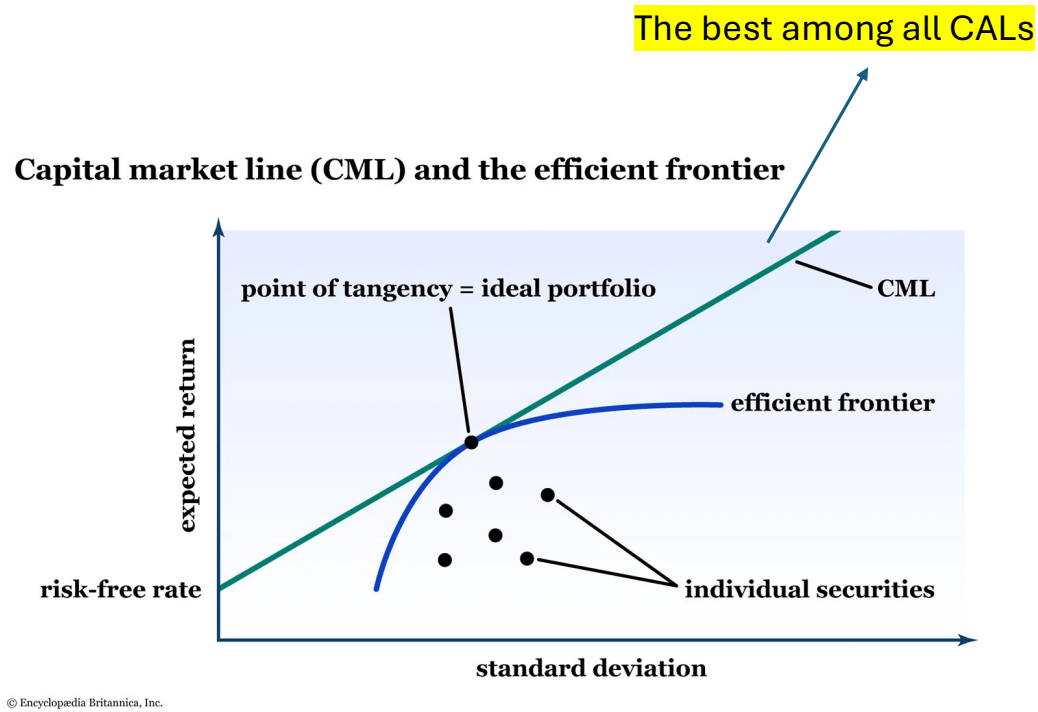
Recap: Modern Portfolio Theory (MPT)

- Imagine applying our 2-asset math to thousands of stocks (the whole market).
- The resulting blue upper-left boundary is the **Efficient Frontier** of all risky assets.
- When we combine the risk-free asset with risky portfolios, the combinations form straight lines.



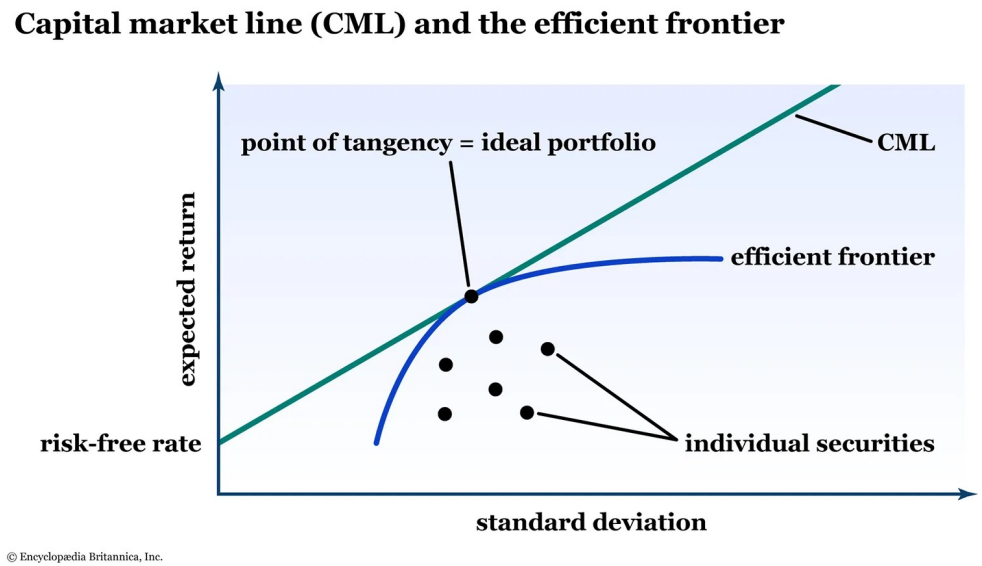
Recap: Modern Portfolio Theory (MPT)

- The green line is tangent to the efficient frontier → this is the **Capital Market Line (CML)**
- CML has the steepest slope among all such lines.
- Slope = excess return per unit of total risk
- In equilibrium, the touching point = tangency portfolio = market portfolio



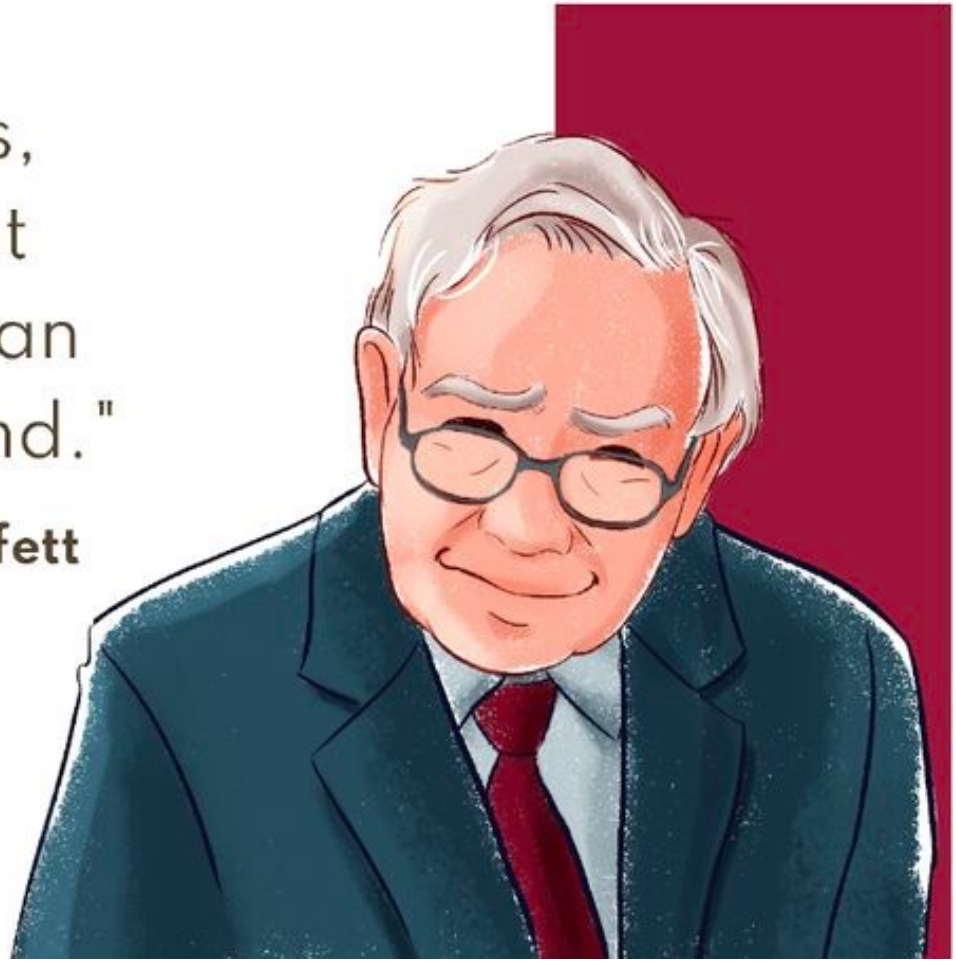
Recap: Modern Portfolio Theory (MPT)

- Therefore, all investors should hold the *same* optimal risky portfolio and combine it with the risk-free asset based on their risk tolerance.
- Conservative investors place more weight on the risk-free asset, while aggressive investors place more weight on the market portfolio.



"For most investors, the best investment strategy is to own an S&P 500 index fund."

- Warren Buffett



My money, I should add, is where my mouth is: What I advise here is essentially identical to certain instructions I've laid out in my will. One bequest provides that cash will be delivered to a trustee for my wife's benefit. (I have to use cash for individual bequests, because *all* of my Berkshire shares will be fully distributed to certain philanthropic organizations over the ten years following the closing of my estate.) My advice to the trustee could not be more simple: Put 10% of the cash in short-term government bonds and 90% in a very low-cost S&P 500 index fund. (I suggest Vanguard's.) I believe the trust's long-term results from this policy will be superior to those attained by most investors – whether pension funds, institutions or individuals – who employ high-fee managers.

Source: Warren Buffett, "2013 Letter to Berkshire Shareholders," Berkshire Hathaway, (https://www.berkshirehathaway.com/letters/2013ltr.pdf)

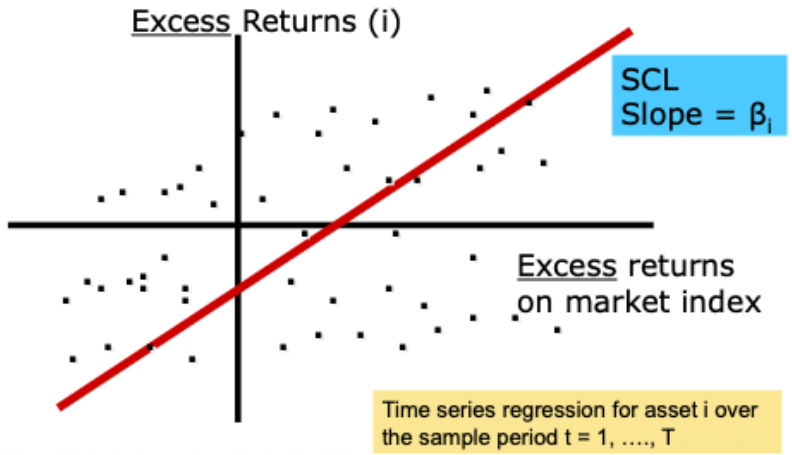
Recap: The Capital Asset Pricing Model (CAPM)

- MPT tells us how to build a giant portfolio. But what if you are a finance manager at Microsoft trying to figure out your **Cost of Capital** for a new project?
- In the 1960s, **William Sharpe** built on Markowitz's MPT and developed the CAPM.
- Total risk (σ) is actually made of two parts:

1. Idiosyncratic risk (firm-specific): A CEO scandal, a factory fore, a failed product.
2. Systematic risk (β, market risk): Recessions, inflation, global pandemics.

Recap: The Capital Asset Pricing Model (CAPM)

- Because MPT proves that anyone can eliminate idiosyncratic risk for free just by diversifying (buying the tangency portfolio), investors are ONLY rewarded for bearing systematic risk.
- To accurately price a stock like Microsoft, we measure how Microsoft reacts to market movements.



Time series regression for asset i over the sample period t = 1, ..., T

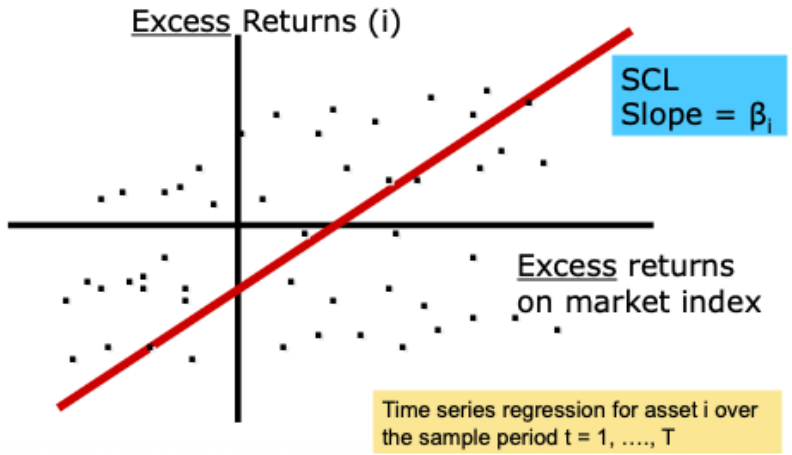
$$R_i^e = \alpha_i + \beta_i R_M^e + e_i$$

Recap: The Capital Asset Pricing Model (CAPM)

- Imagine plotting Microsoft's daily performance against the market.
- X-Axis: Market's excess return
- Y-Axis: Microsoft's excess return
- The **Security Characteristic Line (SCL)** is a "line of best fit" through these points.
- The slope of this line is what we call Beta.

$$\beta = \frac{Cov(i,m)}{Var(m)} = \frac{\sigma_{i,m}}{\sigma_m^2}$$

systematic risk



$$R_i^e = \alpha_i + \beta_i R_M^e + e_i$$

KUNGL. VETENSKAPSAKADEMIEN
THE ROYAL SWEDISH ACADEMY OF SCIENCES

16 October 1990

THIS YEAR'S LAUREATES ARE PIONEERS IN THE THEORY OF FINANCIAL ECONOMICS AND CORPORATE FINANCE

The Royal Swedish Academy of Sciences has decided to award the 1990 Alfred Nobel Memorial Prize in Economic Sciences with one third each, to

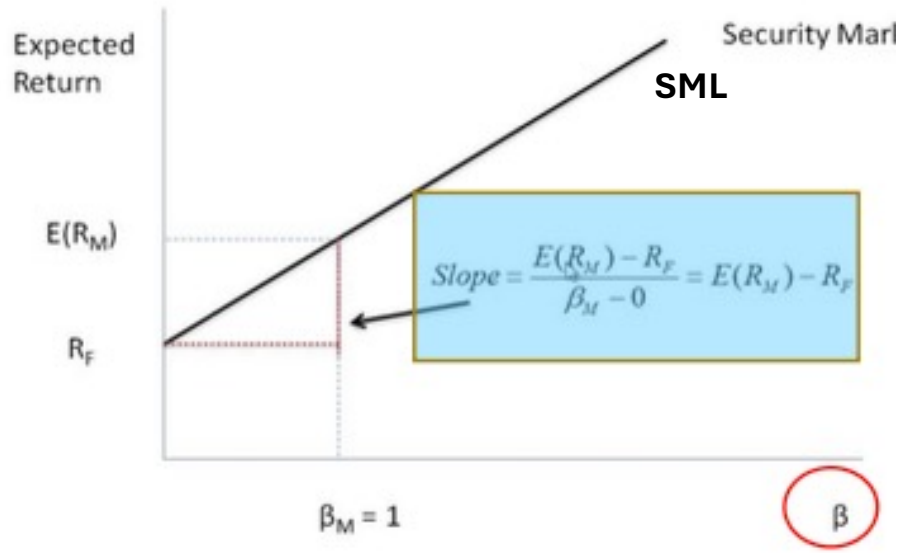
Professor **Harry Markowitz**, City University of New York, USA,
Professor **Merton Miller**, University of Chicago, USA,
Professor **William Sharpe**, Stanford University, USA,

for their pioneering work in the theory of financial economics.

Harry Markowitz is awarded the Prize for having developed the theory of portfolio choice;
William Sharpe, for his contributions to the theory of price formation for financial assets, the so-called, *Capital Asset Pricing Model* (CAPM); and
Merton Miller, for his fundamental contributions to the theory of corporate finance.

Recap: The Capital Asset Pricing Model (CAPM)

- So far, we have used the SCL to estimate a stock's beta (β) from historical data.
- Sharpe's next step was to turn beta into a **pricing model**.
- Under the CAPM, the expected return on an asset depends on:
  - The risk-free rate (R_f)
  - The asset's beta (β_m)
  - The market risk premium ($E(R_m) - R_f$)



Recap: The Capital Asset Pricing Model (CAPM)

the asset i's risk premium (RP)

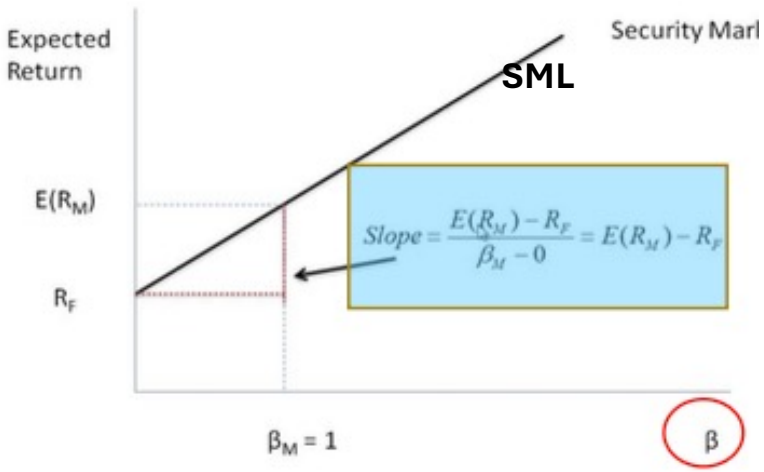
- The CAPM formula is:

$$E(R_i) - R_f = \beta_m(E(R_m) - R_f)$$

Cost of capital

$$E(R_i) = \beta_m(E(R_m) - R_f) + R_f$$

- This equation tells us the return investors should require for bearing systematic risk
- The graphical representation of this relationship is the **Security Market Line (SML).**
- In equilibrium, all correctly priced assets should lie on the SML.



Recap: SCL, SML, CML

- Security characteristic line (SCL) ⟺ market model (i's excess rtn form)
  - is a regression line, plotting the performance of a particular security or portfolio against that of the market portfolio over time
  - The slope is the security's beta, and the intercept is its alpha.
- Security Market Line (SML) ⟺ CAPM (cross-section rtn-beta, i=1,…N)
  - SML is a graphical representation of the CAPM
  - It shows the line that reflects the return on a security in relation to risk. The measure of risk is the security's beta
  - Modern Portfolio Theory mathematically demonstrates that idiosyncratic risk (firm-specific) can be diversified away, leaving only systematic risk
- Capital market line (CML) ⟺ Efficient Frontier for all risky assets (Mkt portfolio is the tangency portfolio). CML is a special case of the Capital Asset Line (CAL)
  - CML is the tangent line drawn from the point of the risk-free asset to the efficient frontier for all risky assets.

How does this one stock behave over time?

beta

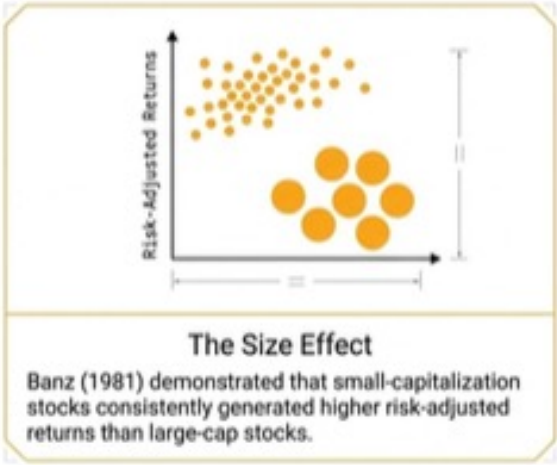
Given a stock's beta, what return should it earn?

assumption

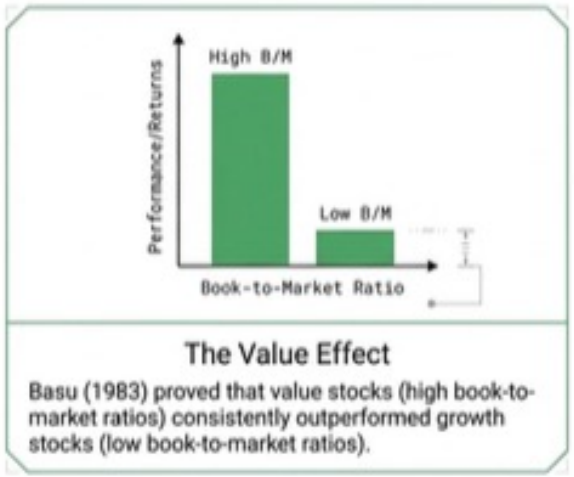
What's the best risk-return trade-off I can get?

Recap: CAPM limitations

- CAPM is elegant, but one beta does not fully explain cross-sectional stock returns (explanatory power: ~70%).
- In practice, stocks with similar betas can still have very different average returns.
- This led researchers to consider **additional risk factors** beyond the market factor.



Anomaly; The market blind spots



5-1

1. Explain what the size effect is.

The size effect is the empirical observation that firms with lower market capitalizations on average have

(1) _____ average returns. (Select from the drop-down menu.)

(1) ○ lower
○ higher

5-1 solution

- The size effect is the empirical observation that firms with lower market capitalization on average have **higher** average returns.

Recap: The Fama–French Three-Factor Model (FF3)

- The FF3 extends the CAPM by **adding** two more risk factors.

1. SMB (Small Minus Big): the return difference between small-cap and large-cap stocks.
2. HML (High Minus Low): the return difference between high book-to-market and low book-to-market stocks.

The model is written as:

$$E(R_i) - \beta_f = \beta_m \times \left(E(R_m) - R_f\right) + \beta_s \times E(SMB) + \beta_h \times E(HML)$$

- FF3 suggests that expected return is influenced not only by market risk, but also by size and value effects.
- This helps explain why some stocks earn returns that CAPM alone cannot fully capture (explanatory power: ~90%).

Recap: The Carhart Four-Factor Model (FFC)

- The Carhart Four-Factor Model (FFC) extends the Fama-French Three-factor Model (FF3) by adding a momentum factor.
- In addition to the market, size, and value factors, Carhart includes:
- WML (Winners Minus Losers) or PR1YR: the return difference between past winners and past losers.
- Formula:

$$E(R_i) - \beta_f = \beta_m \times \left(E(R_m) - R_f\right) + \beta_s \times E(SMB) + \beta_h \times E(HML) + \beta_w \times E(WML)$$

- **Momentum**: the idea is that stocks that have performed well in the recent past often continue to perform well in the near future.

5-2

2. Using the factor beta estimates in Table 1 below and the expected return estimates in Table 2 below, calculate the risk premium of General Electric stock (ticker: GE) using the FFC factor specification. (Annualize your result by multiplying by 12.) GE's CAPM beta over the same time period was 1.32. How does the risk premium compare with the risk premium you would estimate from the CAPM?

Table 1: Estimated Factor Betas

Factor	MSFT	XOM	GE
Mkt	0.965	0.808	1.159
SMB	−0.295	−0.812	−0.496
HML	−0.099	0.196	0.905
PR1YR	−0.089	0.376	−0.196

Table 2: FFC Portfolio Average Monthly Returns, 1927-2012

Factor Portfolio	Average Monthly Return (%)	95% Confidence Band (%)
Mkt − r_f	0.585	±0.34
SMB	0.256	±0.20
HML	0.389	±0.22
PR1YR	0.744	±0.29

WML

The risk premium (monthly) of General Electric stock is _____%. (Round to three decimal places.)

The risk premium (annual) of General Electric stock is _____%. (Round to two decimal places.)

5-2 solution

- The Carhart 4-factor (FFC) model:
  $Risk\ Premium(RP) = \beta_m \times \left(E(Mkt) - R_f\right) + \beta_s \times E(SMB) + \beta_h \times E(HML) + \beta_p \times E(PR1YR)$

GE's FFC beta

GE
1.159
−0.496
0.905
−0.196

Factor Portfolio	Average Monthly Return (%)
Mkt − r_f	0.585
SMB	0.256
HML	0.389
PR1YR	0.744

- $Monthly\ RP = (1.159 \times 0.585) + (-0.496 \times 0.256) + (0.905 \times 0.389) + (-0.196 \times 0.744) = 0.757\%$
- $Annual\ RP = 0.757\% \times \mathbf{12} = 9.09\%$

5-2 (cont.)

GE's CAPM beta over the same time period was 1.32. How does the risk premium compare with the risk premium you would estimate from the CAPM?

The annual RP produced by the CAPM beta is _____%. (Round to two decimal places.)

How does the risk premium compare with the risk premium you would estimate from the CAPM? (Select from the drop-down menu.)

The annual RP produced by the CAPM beta is (1) _____ than the risk premium (annual) of General Electric stock.

(1) ○ lower
○ higher

5-2 solution (cont.)

GE's CAPM beta =1.32

- The CAPM model:

$$RP_{CAPM} = \beta_{CAPM} \times \left(E(Mkt) - r_f\right)$$

- $Monthly\ RP_{CAPM} = 1.32 \times 0.585 = 0.772\%$
- $Annual\ RP_{CAPM} = 0.772\% \times 12 = 9.27\%$
- The annual RP produced by the CAPM beta (9.27%) is higher than the risk premium annual of general electric stock (9.09%).

Factor Portfolio	Average Monthly Return (%)
Mkt − r_f	0.585

GE's CAPM beta over the same time period was 1.32. I

5-3

3. You are currently considering an investment in a project in the energy sector. The investment has the same riskiness as Exxon Mobil stock (ticker: XOM). Using the data in Table 1 and Table 2 below, calculate the cost of capital using the FFC factor specification if the current risk-free rate is 2.7% per year.

Table 1: Estimated Factor Betas

Factor	MSFT	XOM	GE
Mkt	0.965	0.881	1.183
SMB	− 0.295	− 0.892	− 0.475
HML	− 0.099	0.188	0.965
PR1YR	− 0.089	0.385	− 0.200

Table 2: FFC Portfolio Average Monthly Returns, 1927-2012

Factor Portfolio	Average Monthly Return (%)	95% Confidence Band (%)
Mkt − r_f	0.601	± 0.34
SMB	0.253	± 0.20
HML	0.374	± 0.22
PR1YR	0.757	± 0.29

The cost of capital using the FFC factor specification is _____%. (Round to two decimal places.)

5-3 solution

- Step 1: The risk premium calculation:

$$E(R_i) - r_f = Risk\ Premium(RP)$$
$$= \beta_m \times \left(E(Mkt) - r_f\right) + \beta_s \times E(SMB) + \beta_h \times E(HML)$$
$$+ \beta_p \times E(PR1YR)$$

- Step 2: The cost of capital calculation:

$$E(R_i) = Risk\ Premium(RP) + r_f$$
$$= \beta_m \times \left(E(Mkt) - r_f\right) + \beta_s \times E(SMB) + \beta_h \times E(HML)$$
$$+ \beta_p \times E(PR1YR) + r_f$$

5-3 solution

the current risk-free rate is 2.7% per year.

XOM's FFC beta

XOM	Factor Portfolio	Average Monthly Return (%)
0.881	Mkt − r_f	0.601
− 0.892	SMB	0.253
0.188	HML	0.374
0.385	PR1YR	0.757

- $Monthly\ RP_{XOM} = (0.881 \times 0.601) + (-0.892 \times 0.253) + (0.188 \times 0.374) + (0.385 \times 0.757) = 0.666\%$
- $Annual\ RP_{XOM} = 0.666\% \times \mathbf{12} = 7.99\%$
- $E(R_{XOM}) = 7.99\% + 2.7\% = 10.69\%$

5-4

4. You work for Microsoft Corporation (ticker: MSFT), and you are considering whether to develop a new software product. The risk of the investment is the same as the risk of the company.
a. Using the data in Table 1 and Table 2 below, calculate the cost of capital using the FFC factor specification if the current risk-free rate is 3.1% per year.
b. Microsoft's CAPM beta over the same period was 0.84. What cost of capital would you estimate using CAPM?

Table 1: Estimated Factor Betas

Factor	MSFT	XOM	GE
Mkt	0.992	0.879	1.183
SMB	−0.266	−0.807	−0.475
HML	−0.092	0.196	0.965
PR1YR	−0.082	0.381	−0.200

Table 2: FFC Portfolio Average Monthly Returns, 1927-2012

Factor Portfolio	Average Monthly Return (%)	95% Confidence Band (%)
Mkt − r_f	0.591	±0.34
SMB	0.245	±0.20
HML	0.385	±0.22
PR1YR	0.649	±0.29

a. Using the data in Table 1 and Table 2 below, calculate the cost of capital using the FFC factor specification if the current risk-free rate is 3.1% per year.

The cost of capital using the FFC factor specification is _____%. (Round to two decimal places.)

b. Microsoft's CAPM beta over the same period was 0.84. What cost of capital would you estimate using CAPM?

The annual RP produced by the CAPM beta is _____%. (Round to two decimal places.)

5-4 solution

a.

- The cost of capital calculation:

$$E(R_i) = Risk\ Premium(RP) + r_f$$
$$= \beta_m \times \left(E(Mkt) - r_f\right) + \beta_s \times E(SMB) + \beta_h \times E(HML)$$
$$+ \beta_p \times E(PR1YR) + r_f$$

5-4 solution

- $Monthly\ RP_{MSFT} = (0.992 \times 0.591) + (-0.266 \times 0.245) + (-0.092 \times 0.385) + (-0.082 \times 0.649) = 0.432\%$
- $Annual\ RP_{MSFT} = 0.432\% \times 12 = 5.19\%$
- $E(R_{MSFT}) = 5.19\% + 3.1\% = 8.29\%$

a. Using the data in Table 1 and Table 2 below, calculate the cost of capital using the FFC factor specification if the current risk-free rate is 3.1% per year.

MSFT's FFC beta

MSFT
0.992
− 0.266
− 0.092
− 0.082

Factor Portfolio	Average Monthly Return (%)
Mkt − r_f	0.591
SMB	0.245
HML	0.385
PR1YR	0.649

5-4 solution

Factor Portfolio	Average Monthly Return (%)
Mkt − r_f	0.591

b. Microsoft's CAPM beta over the same period was 0.84.

b.

- The CAPM model:

$$E(R_i) - r_f = RP_{CAPM} = \beta_{CAPM} \times \left(E(R_m) - r_f\right)$$

- The cost of capital calculation:

$$E(R_i) = RP_{CAPM} + r_f = \beta_{CAPM} \times \left(E(R_m) - r_f\right) + r_f$$

- $Annual\ RP_{MSFT} = (0.84 \times 0.591) \times 12 = 5.96\%$
- $E(R_{MSFT}) = 5.96\% + 3.1\% = 9.06\%$